# Introduction to Neural Networks

Neural networks are computational models inspired by the human brain. They consist of interconnected nodes that process data, learn complex patterns, and make predictions to solve advanced problems.

## Questions & Answers

**Q1: What is an artificial neural network?**

A1: It is a computing system designed to simulate the way the human brain analyzes and processes information.

**Q2: What are the three main types of layers in a neural network?**

A2: The three main types are the input layer, hidden layers, and the output layer.

**Q3: What is the purpose of an activation function?**

A3: It introduces non-linear properties to the network, allowing it to learn complex data patterns.

**Q4: What is deep learning?**

A4: Deep learning is a subset of machine learning that utilizes deep neural networks with many hidden layers.

**Q5: What is backpropagation used for?**

A5: It is an algorithm used to calculate the gradient of the error function, helping the network update its weights and learn.